SmartRoutes — Intelligent Traffic Management & Route Optimization

Domain 1: Full Stack Web Development (PS 01) | Live Localhost: http://localhost:5000

Executive Summary for Judges: SmartRoutes satisfies all core deliverables of Problem Statement 01: allowing users to input source and destination (with current location locked by default inside a **500m blue accuracy radius**), calculating and displaying **2 to 3 route options** with the fastest in **Solid Blue** and alternatives in **Yellow**, comparing shortest vs. fastest distance and travel time (in hours & minutes), displaying **persistent live road incidents** at all times, providing a **citizen incident reporting modal** with automatic alternate route diversions, and offering a **dual-dashboard architecture** with a secure **Administration portal** requiring login to review and moderate incidents.

Core Feature Implementation Checklist

Core Deliverable	System Implementation	Judge Demonstration Flow
500m Blue Accuracy Circle	L.circle([lat, lng], { radius: 500 }) rendered around GPS origin.	Visible immediately upon launch centered at Rani Chennamma Hub.
Source & Destination Inputs	Default origin auto-filled with current location; destination selectable via text or quick chips.	Type any city or click chips (Dharwad, Bagalkot, Hubballi, Airport) to trigger instant routing.
2-3 Routes Displayed	Fastest route rendered in **Solid Blue**; Shortest and Bypass routes in **Yellow**.	Routes rendered on map with interactive hover and click highlighting.
Compare Shortest vs Fastest	Displays travel times formatted in hours/minutes (e.g. 1h 14m vs 1h 32m) and distance in km.	Green comparison box shows exact minutes and distance saved by selecting fastest path.
Persistent Live Incidents	Active roadblocks, accidents, and congestion permanently plotted on map at all times.	Clicking an incident marker zooms to the hazard and reveals details and severity.
Citizen Incident Reporting	Modal allowing drivers to submit accidents, roadblocks, or congestion with instant diversion.	Click ■ **Report Incident**, submit details, and watch the map reroute traffic immediately.
Dual Dashboard (Admin Login)	**Client Dashboard** for public navigation; **Administration Dashboard** with login gate.	Click ■ **Administration**, log in with admin / admin123 to review and moderate reports.

1-Minute Judge Walkthrough Script

1. Client Map: Point out the 500m blue accuracy circle locked around your current location.
2. Route Comparison: Click **Dharwad**. Show the **Fastest Route in Blue** and **2 Alternative Routes in Yellow**, with distances and travel times in hours and minutes.
3. Incident Reporting: Click ■ **Report Incident**, submit an accident report, and show judges how the system immediately flags the road and recommends an alternate bypass.
4. Admin Portal: Switch to ■ **Administration**, sign in with admin / admin123, and show the moderation queue where administrators review, verify, or close roads in real time.